Password Strength Checker

Python source code — Cybersecurity Mini Project

```
import re
import getpass


def check_password(password):
    score = 0
    suggestions = []

    # Check password length
    if len(password) >= 8:
        score += 1
    else:
        suggestions.append("Use at least 8 characters.")

    # Check uppercase letter
    if re.search(r"[A-Z]", password):
        score += 1
    else:
        suggestions.append("Add an uppercase letter.")

    # Check lowercase letter
    if re.search(r"[a-z]", password):
        score += 1
    else:
        suggestions.append("Add a lowercase letter.")

    # Check number
    if re.search(r"[0-9]", password):
        score += 1
    else:
        suggestions.append("Add a number.")

    # Check special character
    if re.search(r"[^A-Za-z0-9]", password):
        score += 1
    else:
        suggestions.append("Add a special character.")

    # Determine password strength
    if score <= 2:
        strength = "WEAK"
    elif score <= 4:
        strength = "MEDIUM"
    else:
        strength = "STRONG"

    print("\\nPassword Strength:", strength)
    print("Score:", score, "/ 5")

    if suggestions:
        print("\\nSuggestions:")
        for item in suggestions:
            print("-", item)
    else:
        print("Good! Your password meets all five checks.")


print("PASSWORD STRENGTH CHECKER")
password = getpass.getpass("Enter a password: ")
check_password(password)
```

How to run: Save the code as *password_checker.py*, then run *python password_checker.py* in a terminal. The password input is hidden in supported terminals.

Note: Use a made-up test password for demonstrations. This educational checker evaluates basic rules only; it does not guarantee that a password is secure.